Telecom Customer Churn Prediction

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3,333

Customers Analysed

14.5%

Annual Churn Rate

5 Models

Tested & Compared

Business Understanding

Setting the stage — why churn matters and what we set out to solve

The Opportunity

- Acquiring a new customer costs 5-7× more than retaining one
- Even 1% churn reduction = meaningful recurring revenue gain
- Proactive outreach is far cheaper than win-back campaigns

The Challenge

- 14.5% of customers leave each year (~483 annually)
- We only know a customer churned after it's too late
- No system exists to identify at-risk customers early

Our Goal

- Build an AI model to score every customer weekly
- Flag high-risk customers before they decide to leave
- Give the retention team a prioritised call list

Data Understanding

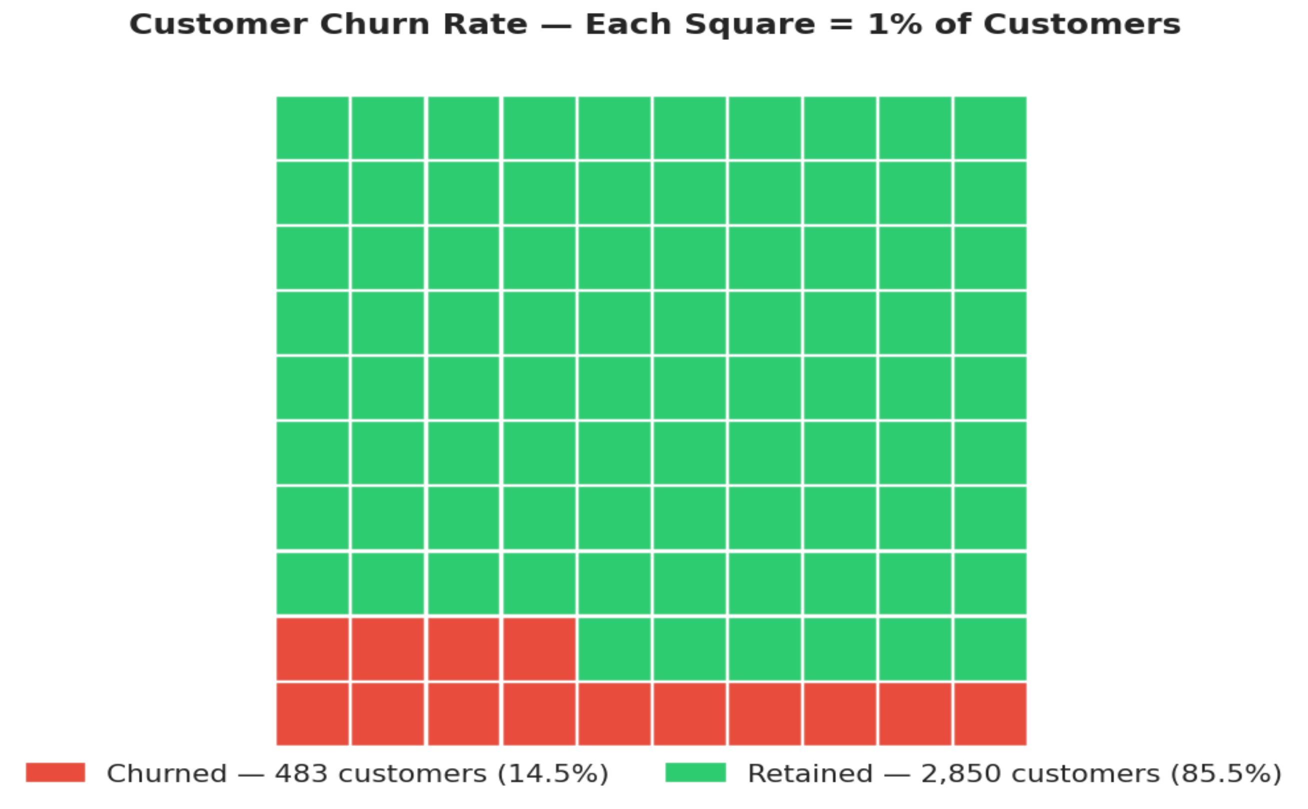
3,333 customers · 21 features · one clear target: did they leave?

3,333
Total Customers

483
Churned (14.5%)

21
Features Analysed

0
Missing Values



Data Preparation

Cleaning and shaping the dataset — what was kept, dropped, and why

7 Columns Dropped

phone number / state / area code
Unique identifiers — no predictive signal

4 charge columns
Pearson r = 1.00 with minute columns — perfectly collinear, add no new information

21 features → 14 features

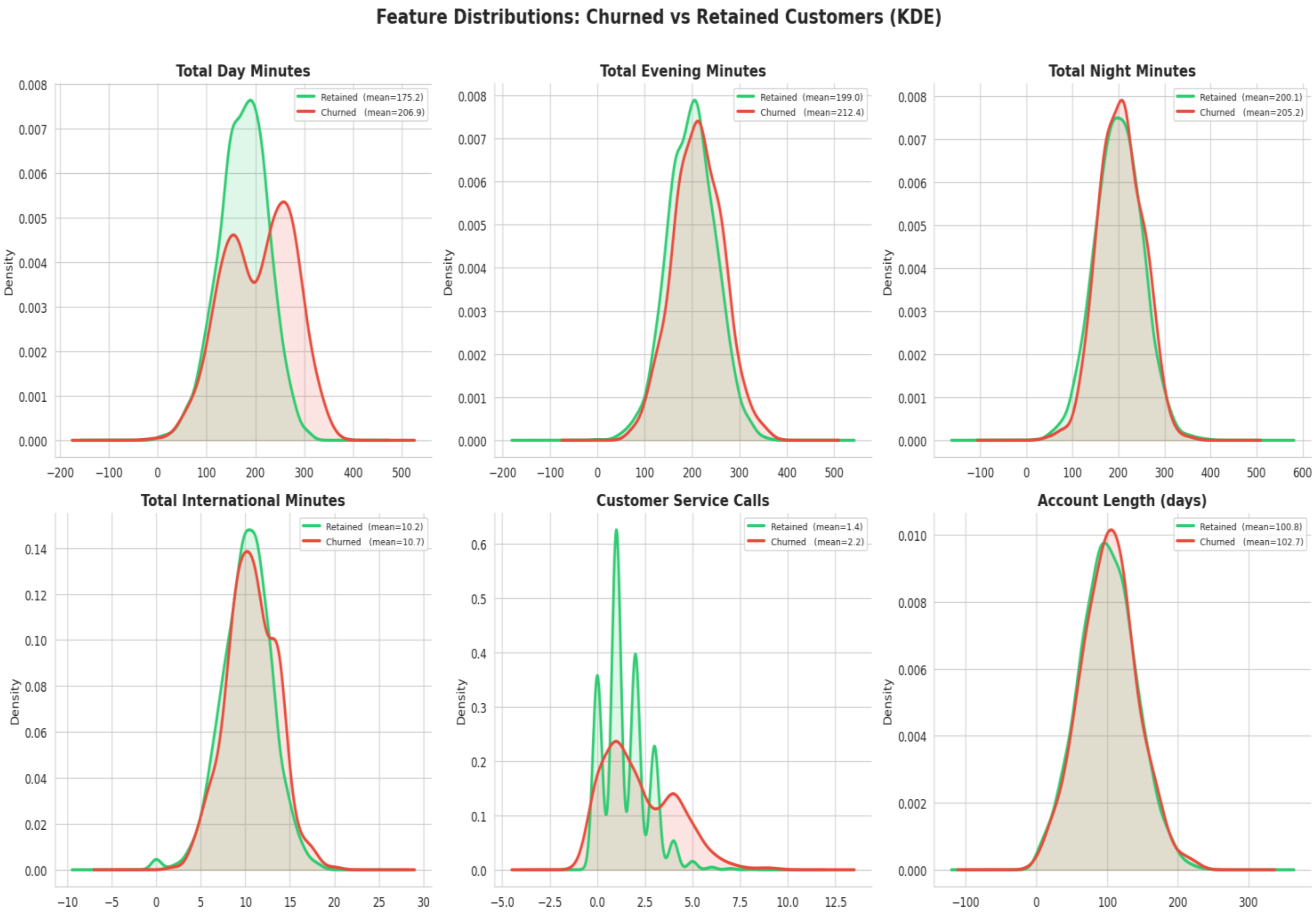
Encoding, Scaling & Split

Binary encoding
international plan, voice mail plan, churn converted to 0/1 integers

StandardScaler
Fitted on train only — prevents test data leaking into model training

80/20 stratified split
2,666 train / 667 test — 14.5% churn rate preserved in both sets

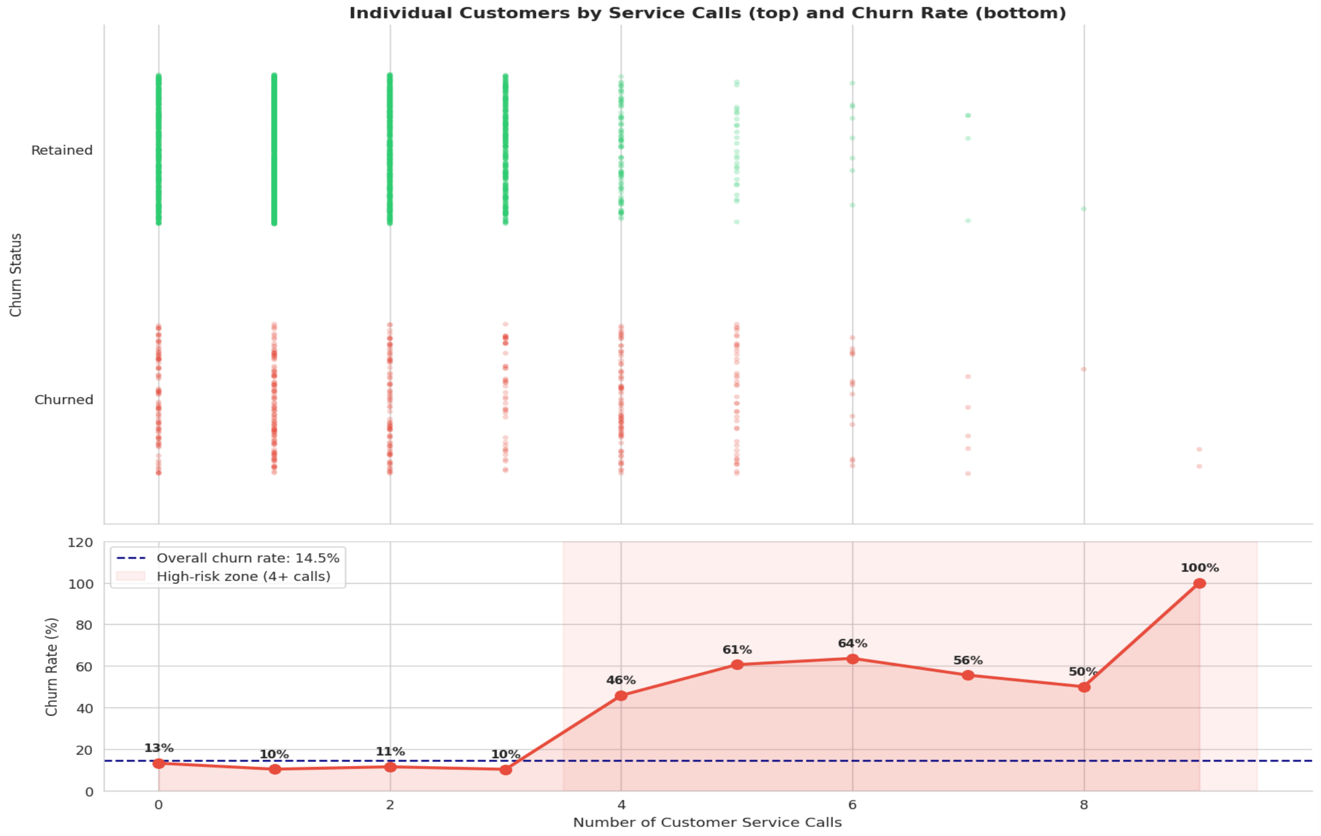
Zero missing values — no imputation needed



Feature Distributions: Churned vs Retained Customers (KDE)
Total Day Minutes
Retained (mean=175.2)
Churned (mean=206.9)
Total Evening Minutes
Retained (mean=199.0)
Churned (mean=212.4)
Total Night Minutes
Retained (mean=200.1)
Churned (mean=205.2)
Total International Minutes
Retained (mean=10.2)
Churned (mean=10.7)
Customer Service Calls
Retained (mean=1.4)
Churned (mean=2.2)
Account Length (days)
Retained (mean=100.8)
Churned (mean=102.7)
Density

Data Finding — The Service Call Signal

The single strongest early warning sign of a customer about to leave



Key Insight

~11%

Churn rate for customers with 0-3 service calls

46-100%

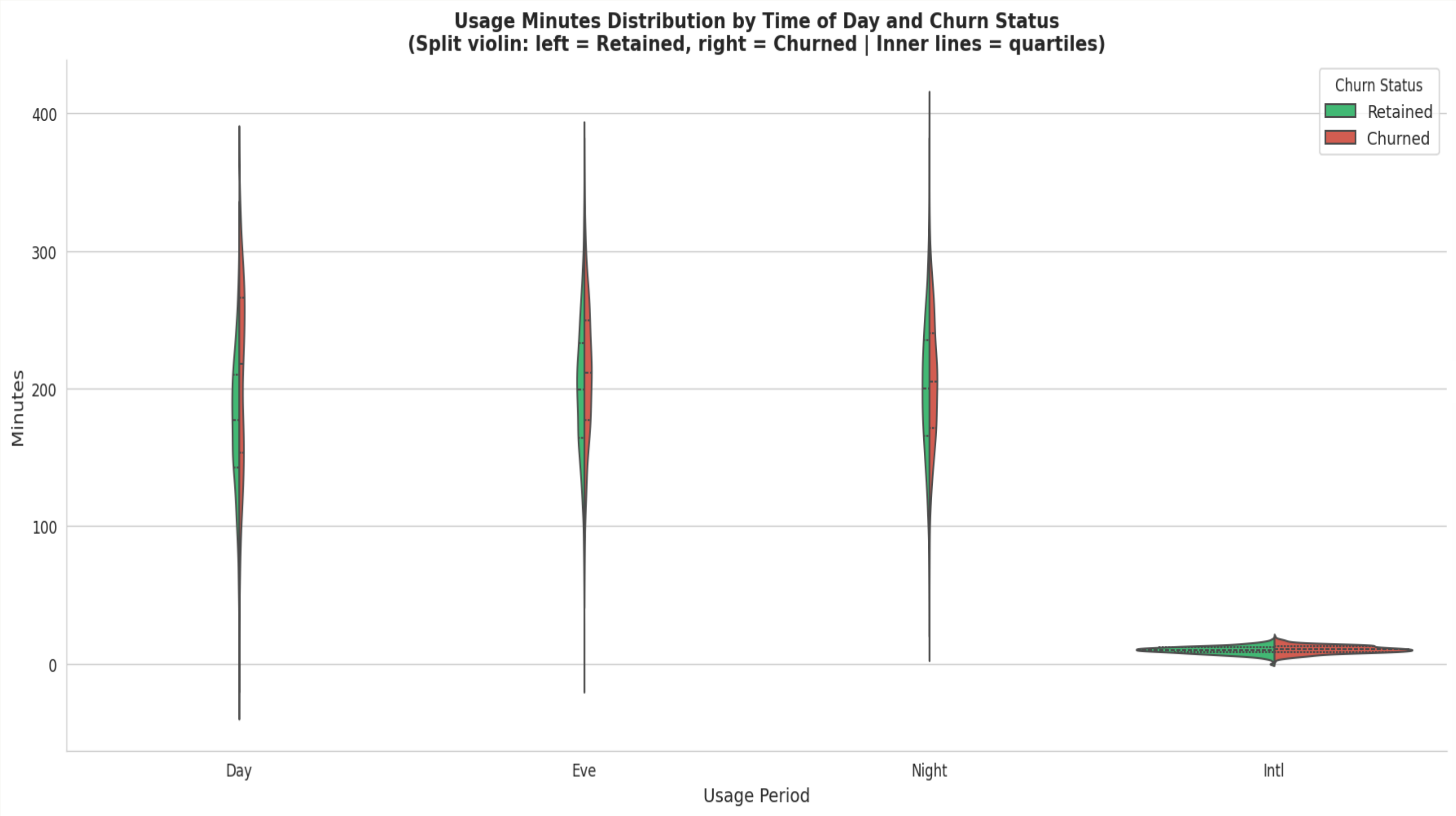
Churn rate for customers with 4+ service calls

Business Implication

After a customer's 3rd call, route them immediately to a senior retention agent — before they decide to leave.

Data Finding — Usage Patterns Drive Churn

Heavy daytime callers face higher bills — and are more likely to leave



Day Minutes

Churners avg **207 min/day** vs 175 for retained — higher usage = higher bills = price pressure

Night & Evening

Night and evening minutes show very similar distributions between churned and retained — these are weaker predictors.

Modeling Approach — Five Models Tested

We tried increasingly powerful approaches until we found the best one

M1

Logistic Regression (Baseline)

Simple but only 25% recall — missed most churners

M2

Logistic Regression (Tuned)

Better: 74% recall, but poor at precision

M3

Decision Tree (Baseline)

Memorised training data — overfit badly

M4

Decision Tree (Tuned)

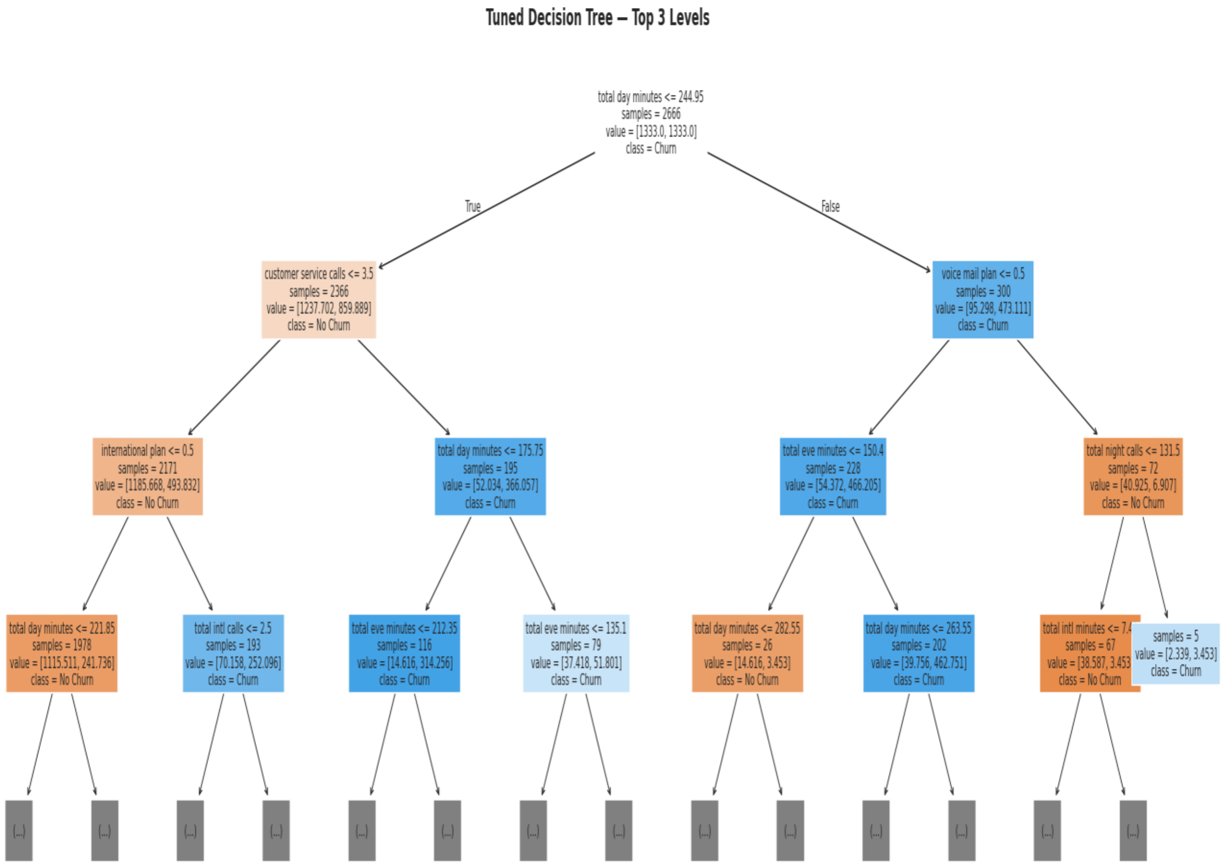
74% recall, better balance — but still overfit

M5

Random Forest ★ FINAL

Best generalisation, highest AUC (0.88), most reliable

Model - Decision tree(Tuned)

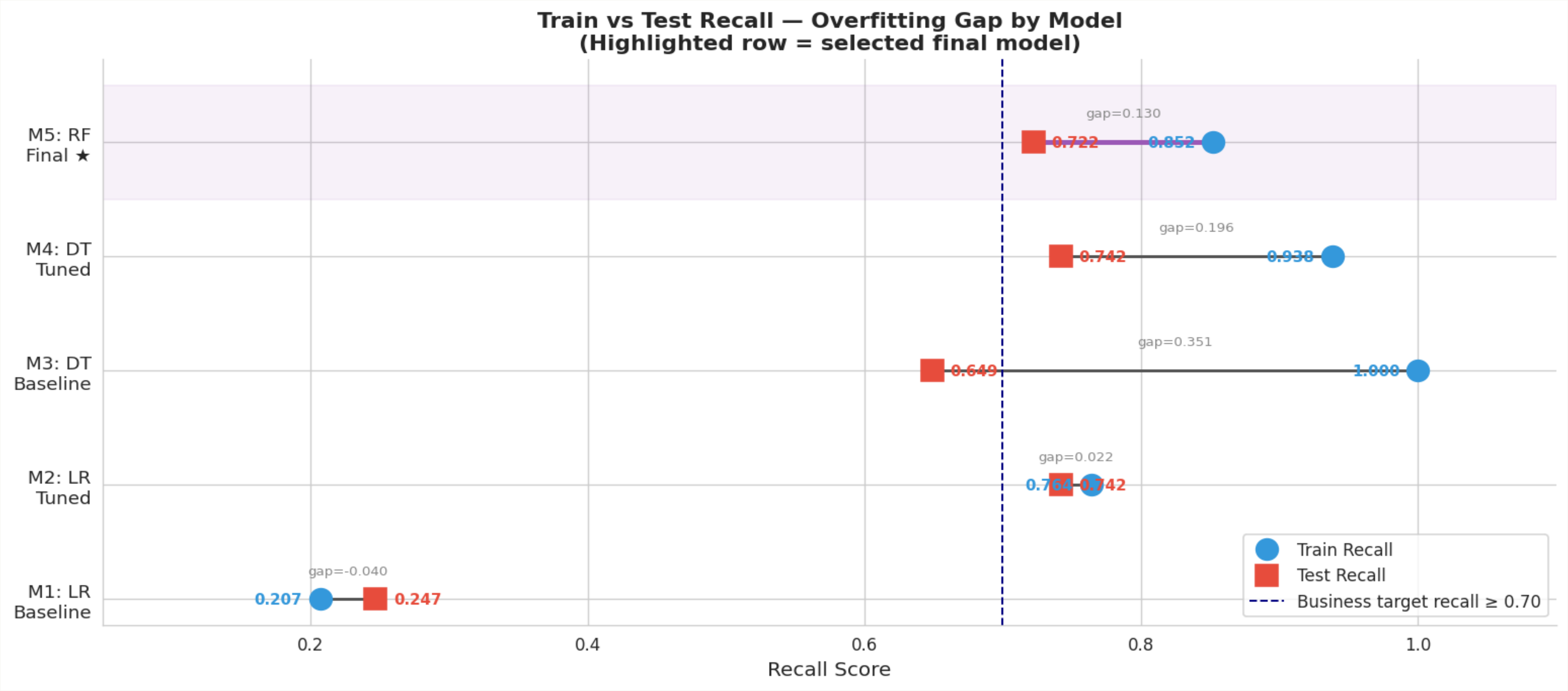


Interpretation
Overfitting gap shrinks from 0.351 to 0.196. Test recall rises to 74.2% matching M2, but F1 is better (0.673 vs 0.474) — same recall with far less collateral. The tree first splits on customer service calls and total day minutes, confirming the EDA insights.

Model Evaluation — Comparing All Five Models

The Random Forest wins on the metrics that matter most for business

Train vs Test Recall — How Much Does Each Model Overfit?



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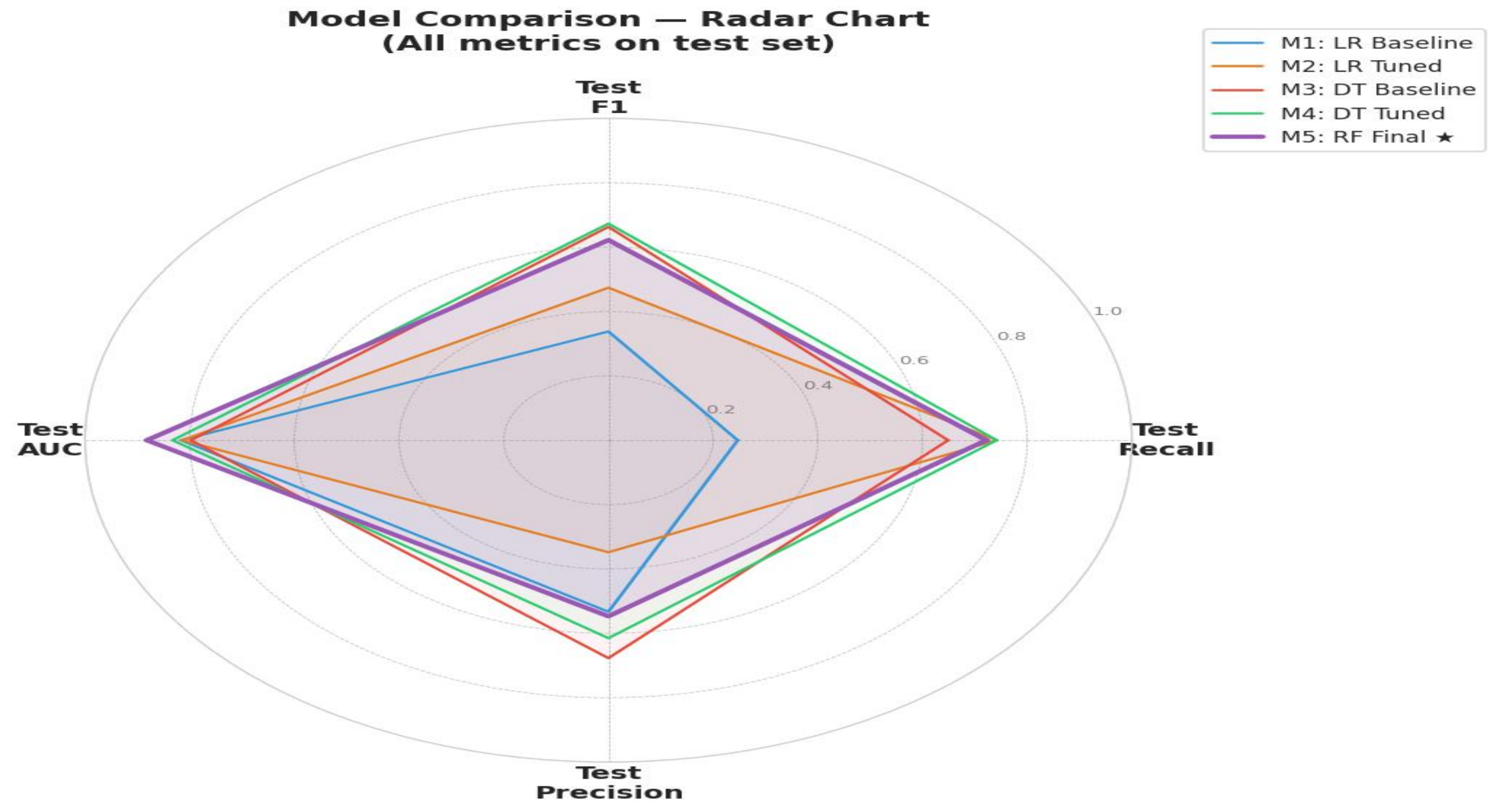
All Metrics at Once — Random Forest Leads



72%

Recall

0.88

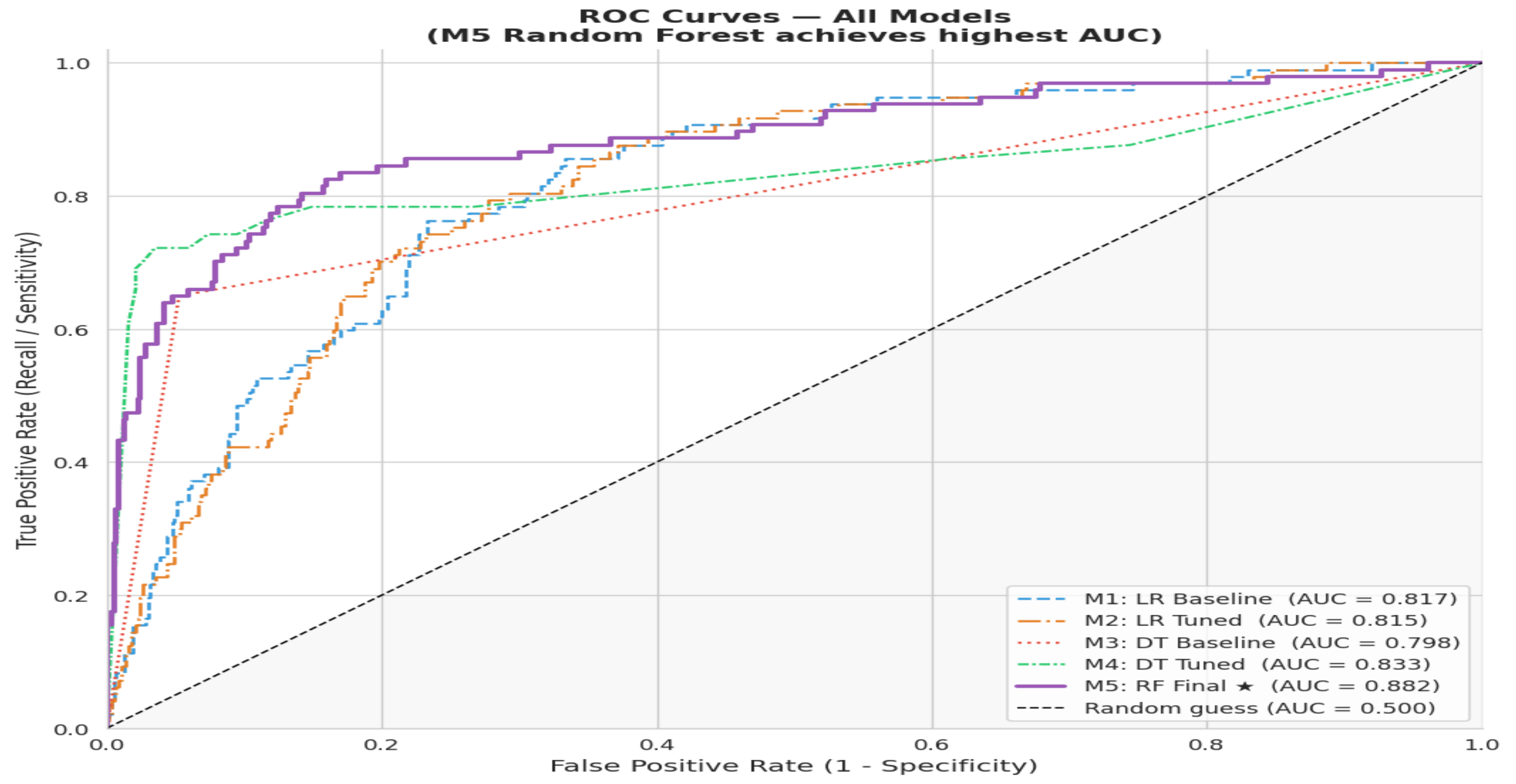
AUC

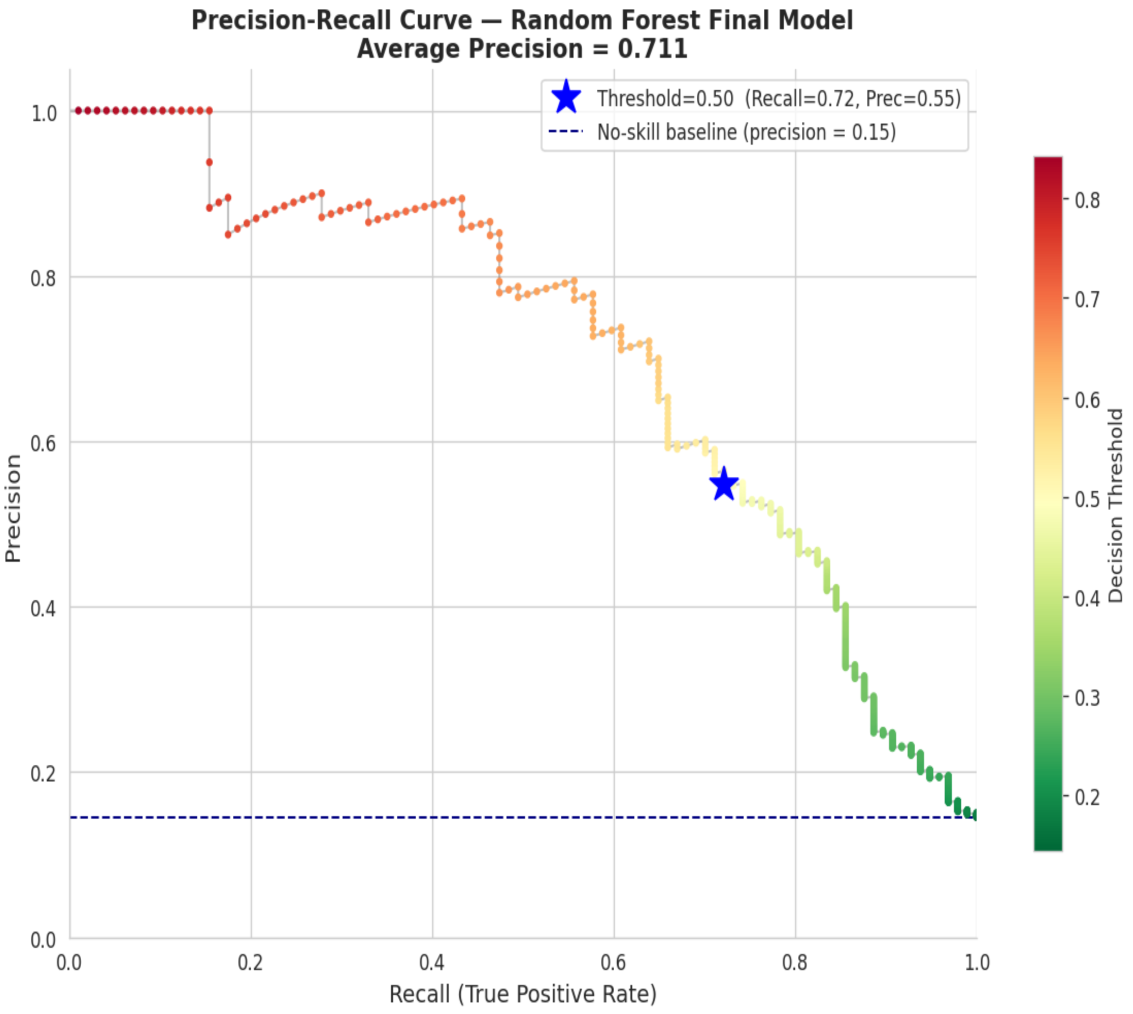
55%

Precision

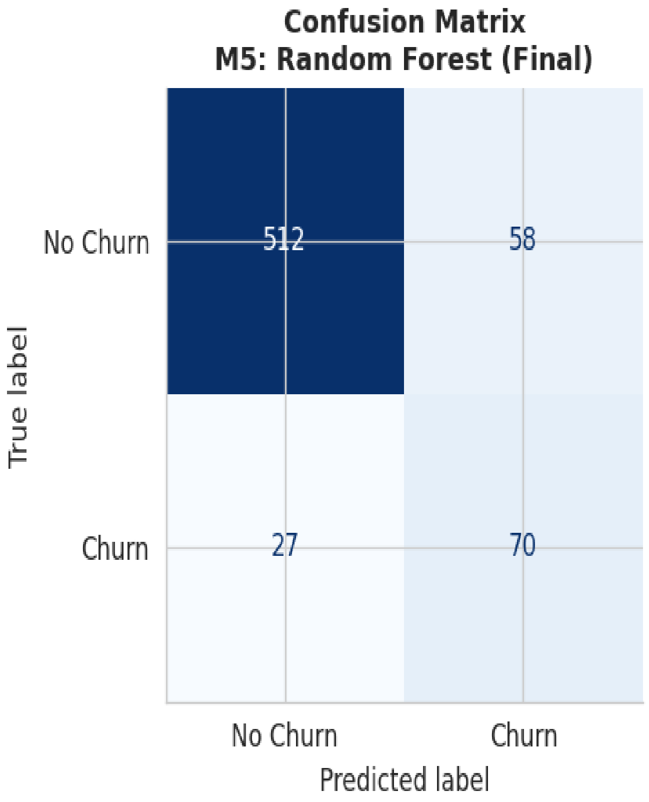
Model Evaluation — What the Numbers Actually Mean

ROC Curve — All Models





Final Model – Confusion Matrix: How Predictions Break Down



Model Evaluation — Explanation

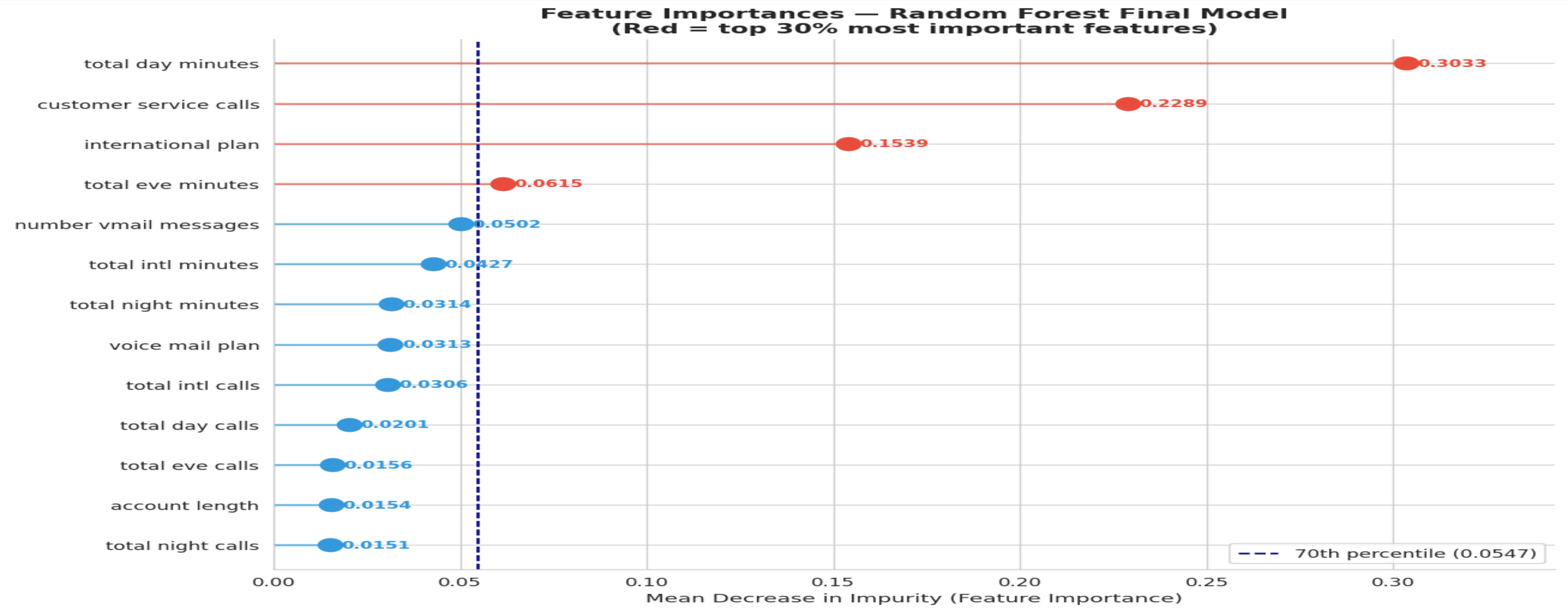
Translating model performance into business outcomes

Plain-English Score Summary

Recall 72%	Of every 100 customers who would leave, we catch 72 and can intervene.
Precision 55%	Of every 100 customers we flag, roughly 55 are genuine churners.
AUC 0.88	Given any churner and non-churner pair, we rank the churner higher 88% of the time.
28% still missed	We can lower this by adjusting the threshold — trading precision for more recall.

What Drives Churn? — Feature Importance

The model reveals which customer behaviours predict departure most strongly



TOP 3 FEATURES — WHAT THEY MEAN FOR US

Total Day Minutes (0.303)

Customer Svc Calls (0.229)

International Plan (0.154)

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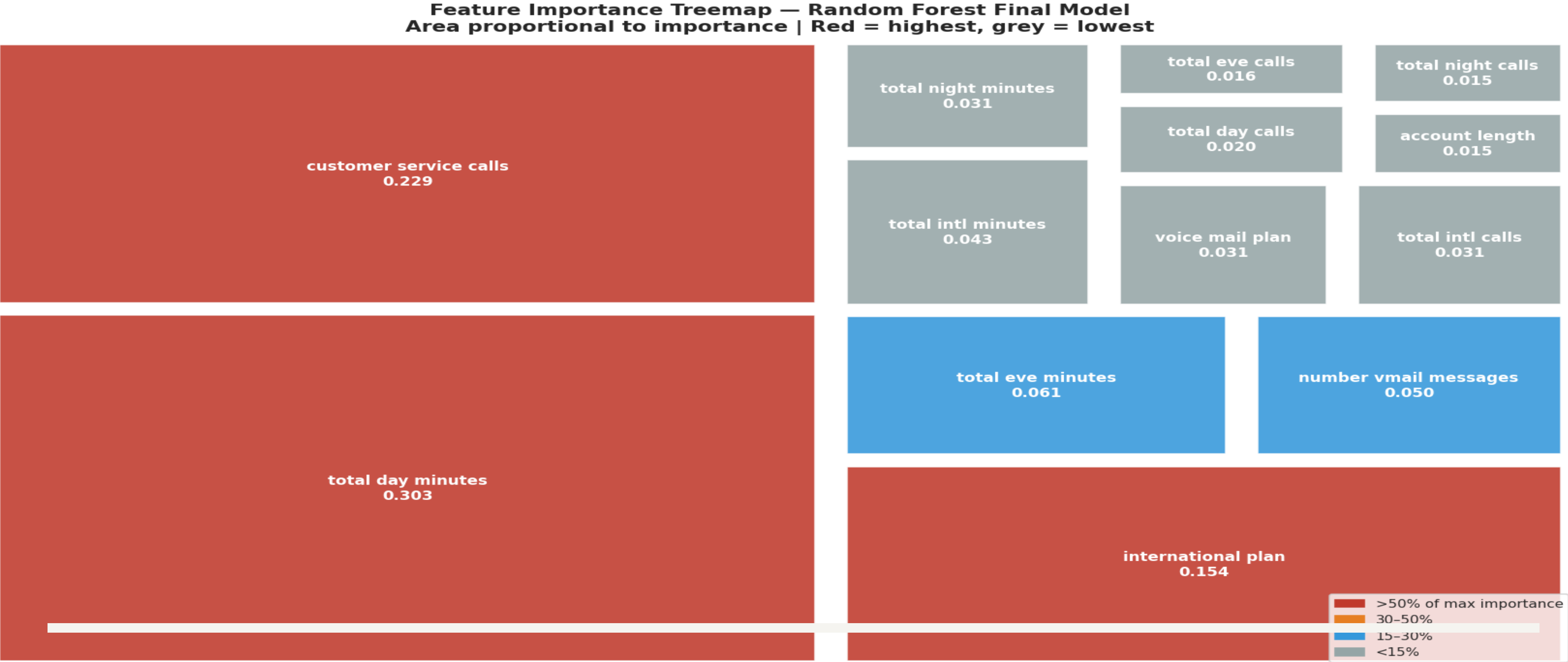

TOP 3 FEATURES — WHAT THEY MEAN FOR US

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Our Three Recommendations

Immediate Action

01

Auto-Flag After the 3rd Service Call

Churn rate jumps from ~11% to 46%+ at 4 calls. Route to a senior retention agent before the 4th call happens. Set up a CRM trigger today.

Within 30 Days

02

Rate Cap Loyalty Offer for Heavy Daytime Users

Flag customers above the 75th percentile in day minutes (>216 min/day). Offer a price guarantee — these are your highest-value and highest-risk customers.

Within 60 Days

03

Audit the International Plan + Offer Voicemail as Incentive

42% of intl plan holders churn — review pricing vs competitors. Separately: voicemail plan holders churn at just 8.7% vs 16.7% — offer it free as a retention incentive.

Next Steps — Deployment Roadmap

Moving from model to measurable business impact

Phase 1
Weeks 1-2

- Deploy weekly scoring pipeline to CRM
- Export priority-ranked at-risk customer list
- Set operating threshold with retention team

Phase 2
Month 1-2

- Run A/B test: hold out control group of high-risk customers
- Measure actual revenue retained vs outreach cost
- Tune threshold based on team capacity

Phase 3
Quarterly

- Retrain model on newest data every quarter
- Add contract type & demographic features
- Target: push recall above 80% with richer data

Model limitation: 27.8% of churners are still missed. Richer data and quarterly retraining will close this gap over time.

Model Limitations & Honest Assessment

What the model gets wrong — and how we plan to close the gaps

28% of churners still missed

Roughly 1 in 4 actual churners are not flagged. Lowering the probability threshold recovers more of these at the cost of more false alarms.

45% false positives among flagged

Nearly half of flagged customers would not have churned. Outreach cost for these must be budgeted into ROI calculations.

No demographic data

Age, income, and contract type are known churn drivers but absent from this dataset

No timestamps

Static snapshot — cannot detect seasonality or model churn as a time-to-event problem

Needs quarterly retraining

Pricing changes and market shifts erode accuracy over time — model must be refreshed regularly

Thank You

Questions & Discussion

72%	0.88	3	483
Churners Caught	Model AUC	Actions to Start	At-Risk / Year

Data Science Team · Moringa School